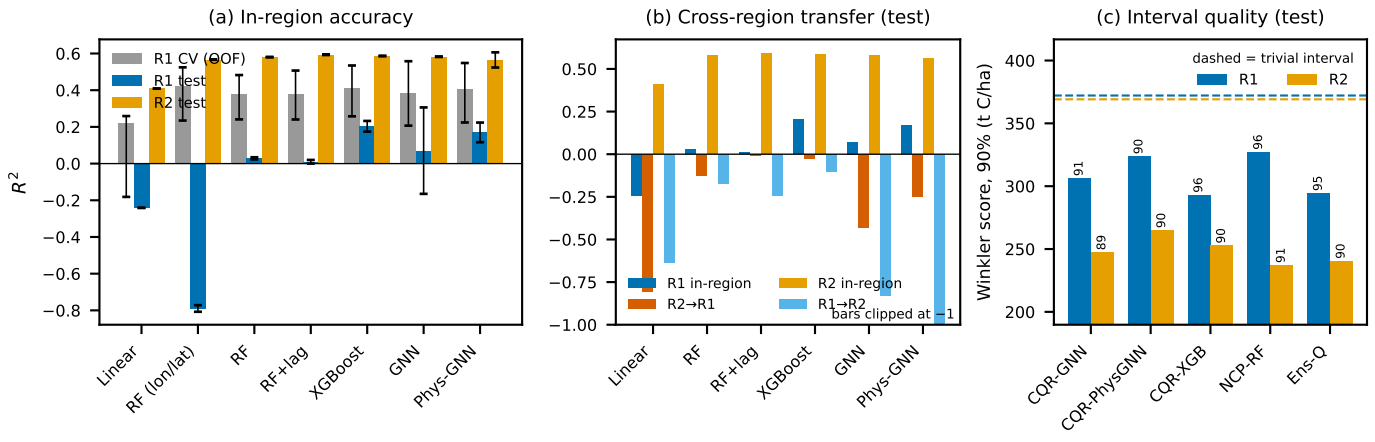
(a) In-region accuracy
R1 CV (OOF)
R1 test
R2 test
R^2
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
Linear
RF (lon/lat)
RF
RF+lag
XGBoost
GNN
Phys-GNN
(b) Cross-region transfer (test)
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
R1 in-region
R2→R1
R2 in-region
R1→R2
bars clipped at −1
Linear
RF
RF+lag
XGBoost
GNN
Phys-GNN
(c) Interval quality (test)
dashed = trivial interval
R1
R2
Winkler score, 90% (t C/ha)
400
350
300
250
200
91
89
90
90
96
90
96
91
95
90
CQR-GNN
CQR-PhysGNN
CQR-XGB
NCP-RF
Ens-Q